

Investigation of Classical Frame Calculations in an NMR Learning Model

Tight advisor overview — ERECTOR SET (scaffold for the ~1 h build)

Shape = the précis, less stuff. Stage 1 reports the genuinely interesting ridge + the distribution-in-space, stats described properly. Stages 2–3 trimmed to essentials. [source] tags mark where each beam's content is staged. Numbers below are vetted (the in-protein truth-vet PASSED them) unless tagged [confirm]. Voice is yours; this is girders.

The question (2 sentences, trimmed)

- Can the NMR shielding **tensor** of a nucleus be accounted for, from **protein geometry alone**, by classical physics kernels (ring current, EFG-from-charge, bond anisotropy, H-bond) calibrated against DFT?
 - Aim is **physics explanation, not a predictor**: *which* geometric mechanisms carry the signal, and *how much*. T2 (the angular part) is kept end-to-end.
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Stage 1 — Static mutant shielding (the reported centerpiece)

Direct question: is the kernel set rich enough to carry the signal across many proteins — and *where in the geometry* does the signal live?

Dataset (one line): 720 WT+ALA AlphaFold pairs, mean pLDDT ≥ 95 (AFDB-confirmed, 226 taxa, ~87% archaeal); target = WT-ALA static DFT $\Delta\sigma$ per matched atom. (*The same 720 DFT serves twice: as the WT-ALA **delta** here, and later as **absolute** shielding to train Model 1.*) [orcas-720, alphafold-accessions]

A. The ridge — what it captures

- per-element, per-atom-type ridge over the **126-kernel T2 layout**.
- **$R^2 = 0.818$** on the 110-protein fair set (69K atoms); **holds at 0.718** across all 720 / 446K atoms — the drop is cross-protein generalisation, not a physics failure. [LEARN_STAGE1]
- backbone full-model R^2 (total |T2|): **0.775 (O) $\rightarrow$ 0.902 (HA)**; total-iso 0.374 (O) $\rightarrow$ 0.950 (HA). [in-protein-stats]
- all-atom medians: total-iso **0.852**, total |T2| **0.812** (242 strata). [in-protein-stats]
- “nitrogen is hard” is a **pooling artifact**: backbone-N ≈ 0.39 vs sidechain-N ≈ 0.89 (2nd-easiest type). [LEARN_STAGE1]
- **effective angular dimensions differ by element**: **H ≈ 20 , C ≈ 6 , N ≈ 3 , O ≈ 12** — the spread is the story. [LEARN_STAGE1]

B. Distribution in space — where the signal sits

- local conformation (ϕ/ψ): all-local T0 CA/N/O = 0.649 / 0.460 / 0.429; O is **ψ -led** (0.304); all-local T2 CA/N/O = 0.547 / 0.411 / 0.318. [in-protein-stats: dihedral]
- local-geometry / “geometry-cloud” probes: signal organises by **local geometric configuration**, selected by geometry not atom-name. [positional summary; topology-workup]
- backbone ϕ/ψ angle-contributor maps: signed z-statistic patterns (direction, not magnitude). [in-protein-stats: OPUS packet]
- **contributors** (who carries it): ring-current scalar leads **T0** (biot_savart 0.734, haigh_mallion 0.721, ring_susceptibility 0.723); q/r^3 + EFG leads **T2** (coulomb 0.763, mopac_coulomb 0.756); ring current also carries T2 (0.662). Rings explicit. [in-protein-stats: contributors]

C. Stats — properly described

- ridge: $\beta = (X^T X + \lambda I)^{-1} X^T Y$; $R^2 = 1 - \text{SSE}/\text{SST}$; per-protein z-normalisation (skip groups < 3). [stats-tests]
 - the test battery: drop-one **mechanism independence**, protein-**blocked CV**, **pairwise independence** (top-20 + named-atom), **weak-but-independent-signal**, **T0/T2 & dia/para split**, **atom-role stratification**, **geometry-cloud probe**, **torsion-pair path**. [stats-tests]
 - R^2 is a **diagnostic** that kernels carry signal — never the optimised metric (model $\neq$ physics).
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Stage 2 — The laws (*précis shape, trimmed*)

- **one classical object** behind the calculators: $\sigma_{ab}(i) \approx \sum_s l_s \cdot D_{ab}(r_{is})$, with $D_{ab} = \partial_a \partial_b (1/\rho) = (3\hat{\rho} - \delta)/\rho^3$. [fig: Model-1 architecture-is-the-equation]
 - within-axis on **1P9J**: charge q/r^3 (0.278) and ring (0.275) recover; the **unified D_{ab} -sum recovers the through-space tensor (0.432)** — a *real* combine, carried by MOPAC-field + McConnell, charge ≈ 0 . [in-protein-stats: Stage 2]
 - honest framing: 1P9J is the **within** instrument; the between-axis is deferred to the **720-WT statics pilot**.
 - grading: **statistical position, not literature-match**. Correlate, do not match.
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Stage 3 — The learning model (*forward arc — the part they'll push*)

- **Ethos flip**: the question changes here — from *which mechanisms carry the signal* (explanation; R^2 a diagnostic) to *do they help predict the measured shift* (utility; **R^2 is the metric**). The explanation goal isn't dropped — Stage 3 tests whether the explained physics **earns its place**.
 - **Model 1 — small shielding emulator**. e3nn equivariant; **trains on the DFT** (720 statics + 1P9J frames). Stages 1-2 set *which features and what structure* it carries (kernel shadows, the D_{ab} combine), not its training data. Predicts σ where there is no DFT. Prototype already runs: held-out 1P9J **T2 $R^2 \approx 0.48$** . [fig: Model-1; capability graph]
 - **Model 2 — big shift predictor (the wild west)**. Eats Model 1 + the extraction firehose on the ~ 656 no-DFT fleet, trained vs **BMRB** shifts. **Firewall**: the DFT-trained Model 1 is frozen; the noisy experiment only ever trains Model 2 — it never corrupts the physics. [fig: Model-2 wild-west]
 - **The payoff — ablation**. Toggle structure quality / atom pLDDT, Model 1 in/out, and the calculators in/out $\rightarrow$ read what each contributes to the real observable. This is where “*does any of this help*” gets answered.
 - arc: **distill $\rightarrow$ transfer $\rightarrow$ test** (1P9J is the calibration case — it carries both DFT and BMRB).
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The arc, in one line

signal $\rightarrow$ equation calibrations (each law + coefficient + confidence) $\rightarrow$ ensemble $\rightarrow$ equivariant transferability pilot.

Figures / panels on hand (*for selection in the hour*)

- fig_overall_flow — **the whole arc, tying it together** (Mermaid) — likely the opening figure
- fig_shielding_model — Model 1, architecture-is-the-equation (TikZ)
- fig_shift_model — Model 2, the wild west (Mermaid)
- interp_1p9j_advisor_graph — capability graph (held-out T2 $R^2 \approx 0.48$)
- dataset panels: taxa_map, plddt_distribution, size histograms

Open beams for the hour: title; which Stage-1 numbers make the cut vs. sit in back-pocket; figure selection (which 2-3 actually go on the desk); how hard to lean on Stage 3 (the advisor will); the one-line framing of the archaeal-set + pLDDT $\geq$ 95 selection.